

CHATAKE INNOWORKS PVT. LTD.

MindForgeAI Internship Programme

Apollo AgriVerse

Engineering an Intelligent Agricultural Ecosystem Through Digital-Twin Soil and Hydrogel Intelligence

Project Abstract & Outline

Project Title	Apollo AgriVerse
Course Title	Engineering Internship Project
Student Name	Sunaina Gaikwad
Student ID	2026IT002
Degree Program	Diploma in Information Technology
Research Area	Artificial Intelligence, Machine Learning & Digital Twin Systems
Employing Organization	Chatake Innworks Pvt. Ltd., Solapur, Maharashtra, India
Date	July 2026

1. Broad Area of Work

Apollo AgriVerse operates at the intersection of Artificial Intelligence, Machine Learning, Digital Twin modelling, and precision agriculture. Rather than treating soil health monitoring and irrigation management as separate, disconnected tools, the project builds a single simulated environment in which soil condition, water retention behaviour, and crop stress signals can be observed, modelled, and predicted together.

The internship narrows this broad vision to two flagship subsystems: Intelligent Soil and Intelligent Hydrogels. Intelligent Soil is treated not as a static growing medium but as a dynamic substrate whose moisture, nutrient (NPK) levels, pH, and microbial stress indicators change continuously and can be modelled mathematically. Intelligent Hydrogels are simulated as adaptive water-retention agents whose release behaviour responds to transpiration and evaporative demand. Together, these two subsystems form the analytical core of a digital twin that mirrors real field conditions inside software.

2. Background

Farming decisions have traditionally relied on manual observation and reactive intervention — irrigation and treatment are applied only after visible signs of crop or soil stress appear. By this stage, yield loss, excess water usage, or nutrient depletion has often already occurred. Existing precision-agriculture tools tend to monitor a single variable in isolation (soil moisture only, or weather only) and rarely connect these readings to a predictive decision layer.

Apollo AgriVerse addresses this gap by proposing a proactive, model-driven alternative. Instead of waiting for a sensor to report a problem, a Digital Twin continuously simulates the expected state of the soil and hydrogel systems, allowing early detection of stress patterns and enabling irrigation or nutrient decisions to be made before physical damage occurs. This internship phase focuses on building the simulated, software-only foundation of that twin — generating realistic synthetic telemetry for soil and hydrogel behaviour and using it to train early decision-support models — before any physical hardware deployment is attempted in later research phases.

3. Objectives & Scope

The project is guided by the following measurable objectives:

- Design a modular Digital Twin engine capable of simulating Intelligent Soil behaviour, including moisture depletion, NPK drift, and pH variation over time.
- Model Intelligent Hydrogel response curves that adjust water release based on simulated transpiration and evaporative demand.
- Generate multi-dimensional synthetic telemetry datasets representing soil and hydrogel state across a simulated field grid.
- Implement and train baseline machine learning models (regression and classification) that convert simulated soil/hydrogel readings into irrigation and stress-alert recommendations.
- Build a lightweight web dashboard that visualises live twin state, predicted stress zones, and irrigation recommendations for a non-technical end user.

Scope boundaries: Within the 4-week internship window, work is confined to the simulated, software layer of the soil and hydrogel subsystems and the analytics built on top of them. The following are explicitly out of scope for this phase: physical deployment of soil or hydrogel sensors on live agricultural land, chemical formulation or manufacture of real hydrogel materials, direct interfacing with physical IoT hardware, and training of deep computer-vision (CNN/DNN) models on real field imagery. These items are reserved for later, multi-month research phases beyond the internship.

4. Plan of Work & References

The 4-week execution timeline is organised into four sequential phases:

Phase	Timeline	Focus Areas & Deliverables
Phase 1: Architecture & Design	Week 1	Literature review, environment setup, mathematical modelling approach for soil moisture/NPK/pH and hydrogel release curves, drafting of this Project Abstract.
Phase 2: Digital Twin Core	Week 2	Development of synthetic telemetry generators for Intelligent Soil and Intelligent Hydrogel states; core twin synchronisation logic.
Phase 3: AI Models & Dashboard	Week 3	Training of regression/classification models for irrigation and stress recommendations; building the interactive dashboard UI.
Phase 4: Assembly & Verification	Week 4	End-to-end integration testing, performance tuning, final report writing, and repository publishing.

Literature References:

1. McKinney, W. (2017). Python for Data Analysis: Data Wrangling with Pandas, NumPy, and IPython (2nd ed.). O'Reilly Media.
2. VanderPlas, J. (2016). Python Data Science Handbook: Essential Tools for Working with Data. O'Reilly Media.
3. Tao, F., Zhang, M., Liu, Y., & Nee, A. Y. C. (2018). Digital Twin Driven Smart Manufacturing. Academic Press.
4. Jones, H. G. (2013). Plants and Microclimate: A Quantitative Approach to Environmental Plant Physiology (3rd ed.). Cambridge University Press.

5. Liakos, K. G., Busato, P., Moshou, D., Pearson, S., & Bochtis, D. (2018). Machine Learning in Agriculture: A Review. *Sensors*, 18(8), 2674.
6. Chatake Innworks Pvt. Ltd. (2026). *MindForgeAI Internal Syllabus: Computational Engineering & Smart Systems*. Solapur, MH.

5. Particulars of the Supervisor

Name	Mr. Akash Shivdas Chatake
Qualification	MTech in AI/ML
Designation	Head Director / Research Lead
Employing Organization	Chatake Innworks Pvt. Ltd. (MindForgeAI Division)
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6. Remarks of the Supervisor

(To be filled by the supervisor upon technical review of the proposed project abstract. Remarks regarding approval, scope adjustments, and engineering standards will be documented here.)

Date of Review	Signature of Student	Signature of Supervisor